PharmaCorp Research Inc.

Clinical Trials Operations and Data Management

Internal Business Description Document

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1. Company Overview and Mission

PharmaCorp Research Inc. is a mid-sized pharmaceutical company specializing in the development of novel therapeutics across multiple therapeutic areas, including **oncology**, [redacted], and [redacted]. Founded in 1995, the company has grown to manage an active portfolio of over 20 clinical trials at any given time, with sites across the United States and partnerships with major academic medical centers and community hospitals.

> ***Our Mission:*** *To advance patient care through rigorous, ethical clinical research that generates high-quality evidence for regulatory approval and clinical practice.*

We conduct **Phase 1 through Phase 4** trials, with a focus on patient safety, data integrity, and regulatory compliance at every stage of the drug development lifecycle.

Company at a Glance

Metric	Value
Founded	1995
Headquarters	Boston, MA
Active Clinical Trials	20+
Therapeutic Areas	4 (Oncology, Cardiovascular, Metabolic, Neurology)
Trial Sites	40+ across the United States
Registered Patients	~300 active participants
Physicians in Registry	50+
Hospital Partners	15

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2. Clinical Trial Operations

2.1 Trial Design and Execution

Clinical trials at PharmaCorp follow a structured lifecycle from protocol development through database lock and regulatory submission. Each trial is assigned a unique trial identifier and is characterized by:

- **Phase** — Phase 1, 2, 3, or 4
- **Therapeutic area** — Oncology, Cardiovascular, Metabolic Disorders, or Neurology
- **Sponsor** — PharmaCorp or a partner organization
- **Target enrollment** — Expected number of participants

Trials are conducted at multiple **trial sites**, each associated with a hospital or research institution. Each site has a designated **principal investigator (PI)** responsible for the conduct of the trial at that location and an enrollment target.

Enrollment Process

Patient enrollment follows a formal workflow:

1. **Screening** — Patients are assessed for eligibility criteria
2. **Informed Consent** — Eligible patients provide written consent
3. **Randomization** — Patients are assigned to a treatment arm (placebo, low dose, medium dose, high dose, or active control)
4. **Monitoring** — Enrollment status is tracked throughout: *screened → enrolled → active → completed / withdrawn / lost to follow-up*

This structure supports enrollment tracking, retention analysis, and regulatory reporting.

2.2 Site and Investigator Management

Trial sites are the physical locations where clinical research is conducted. Each site is linked to a hospital or medical center and to a specific clinical trial.

Site Status Lifecycle:

Status	Description
Active	Site is conducting trial activities
Recruiting	Site is actively enrolling new patients
Closed	Site has completed or terminated participation
Suspended	Site activities temporarily halted

Principal investigators are physicians who hold the clinical and regulatory responsibility for the trial at their site. They must meet sponsor and regulatory requirements for:

- Training and certification
- Clinical experience in the relevant therapeutic area
- Conflict of interest disclosure

PharmaCorp maintains a **physician registry** that includes specialty, license number, and hospital affiliation. Physician participation in multiple trials is tracked to support workload balancing and expertise matching.

2.3 Patient and Visit Management

Patients are the central participants in clinical trials. Each patient record includes:

- **Demographics** — Name, date of birth, gender, contact information
- **Clinical attributes** — Blood type, medical history
- **Insurance** — Plan identifier for billing and compliance (does not affect trial eligibility)

Patient participation is tracked through **visits**. Each visit is associated with an enrollment and has:

Visit Type	Purpose
Screening	Initial eligibility assessment
Baseline	Pre-treatment measurements
Week 4, 8, etc.	Scheduled follow-up assessments
End of Study	Final visit and evaluation
Unscheduled	Clinically necessary off-schedule visit
Follow-up	Post-treatment monitoring

Visit status (*completed, missed, partial, scheduled*) supports protocol adherence monitoring and data completeness reporting.

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3. Regulatory Compliance and Patient Safety

3.1 Consent and Ethical Oversight

Informed consent is a cornerstone of ethical clinical research. All enrollments require a valid consent record.

Consent Types:

- **Initial Informed Consent** — Primary consent before trial participation
- **Assent** — For minors or participants who cannot provide full consent
- **Reconsent** — Required when protocol amendments affect participant rights
- **Withdrawal of Consent** — Participant exercises the right to discontinue

Each consent record includes the consent date, version, signatory, and witness. Version control

ensures participants are enrolled under the correct protocol version and that amendments are properly documented.

3.2 Adverse Event Monitoring

Patient safety is monitored through **adverse event (AE)** reporting. Every adverse event is linked to an enrollment and includes:

Field	Values
Severity	Mild, Moderate, Severe, Life-threatening
Outcome	Recovered, Recovering, Ongoing, Sequelae, Fatal
Relatedness	Related / Not related to investigational treatment

Events are reported by the treating physician or site staff. Aggregation of adverse events by trial, drug, and severity supports:

- Safety signal detection
- Regulatory reporting (DSMB, FDA)
- Risk-benefit assessment

3.3 Lab Results and Clinical Data

Laboratory results are collected at visits to monitor safety and efficacy. Each lab result includes the test name, value, unit, normal range (low and high), and an **abnormal flag**.

Abnormal values trigger:

1. Clinical follow-up
2. Possible dose adjustments
3. Protocol modifications
4. Adverse event reporting (if clinically significant)

Lab data supports pharmacokinetic analysis, safety monitoring, and efficacy endpoints.

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4. Data Management Challenges

4.1 Fragmentation and Integration

Clinical data at PharmaCorp originates from multiple systems:

- **Electronic Data Capture (EDC)** — Trial-specific clinical data
- **Laboratory Information Systems (LIS)** — Lab results and reference ranges
- **Hospital Systems** — Patient demographics and medical history

- **Regulatory Databases** — Submission and compliance records

Patient identifiers, enrollment identifiers, and visit identifiers must be consistently mapped across systems. A unified data model linking patients, trials, enrollments, visits, lab results, adverse events, and consent would enable automated reporting and reduce reconciliation errors.

4.2 Traceability and Auditability

[redacted] to final analysis:

- [redacted]s to
- When and by whom each record was created or modified
- Complete audit trail for all data changes

[redacted]eign key relationships, but the proliferation of spreadsheets and ad-hoc exports increases the risk of orphaned or inconsistent data. Standardizing on a **single source of truth** with clear relationships and lineage would improve audit readiness.

4.3 Patient 360 and Cross-Trial Analysis

A patient may participate in multiple trials over time or have concurrent prescriptions and diagnoses from routine care. A unified **Patient 360** view would combine:

- [redacted]
- Prescriptions
- Diagnoses
- Medical procedures
- Adverse events

This supports safety analysis (drug-drug interactions, comorbidities), recruitment efficiency, and real-world evidence integration.

[redacted]

4.4 Drug Safety and Pharmacovigilance

Drug safety requires correlating adverse events with drugs, both investigational and concomitant. Prescription data captures:

- Drug name, dosage, frequency, and duration
- Prescribing physician
- Start and end dates

A **drug safety view** that aggregates adverse events by drug, severity, and treatment-relatedness supports pharmacovigilance obligations and regulatory reporting.

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5. Data Model Summary

The PharmaCorp clinical trials data model comprises the following entities:

Core Entities

Entity	Description	Approx. Records
Patient	Trial participants	300
Physician	Investigators and prescribers	50
Hospital	Medical centers and research sites	15
Clinical Trial	Sponsored research studies	20
Trial Site	Physical locations for trial conduct	40
Drug	Investigational and reference medications	30
Insurance Plan	Coverage plans for billing compliance	10

Transactional Entities

Entity	Description	Approx. Records
Enrollment	Patient participation in a trial	500
Visit	Scheduled and unscheduled patient visits	1,500
Lab Result	Laboratory test results per visit	2,000
Adverse Event	Safety events linked to enrollments	400
Prescription	Medications prescribed to patients	800
Diagnosis	Clinical diagnoses for patients	600
Medical Procedure	Procedures performed on patients	500
Consent	Informed consent records	500

Total records: ~7,265

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6. Key Performance Indicators

KPI	Target	Current
Enrollment completion rate	≥ 90%	Tracked per trial
Adverse event reporting time	< 24 hours	Monitored continuously
Protocol deviation rate	< 5%	Per site
Data query resolution time	< 7 days	Per trial
Site activation time	< 60 days	From contract to first patient
Lab result turnaround	< 48 hours	Per visit

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7. Appendix: Entity Relationship Overview

Primary Relationships

```
Patient ──(1:N)──► Enrollment
Clinical Trial ──(1:N)──► Trial Site
Trial Site ──(N:1)──► Hospital
Enrollment ──(1:N)──► Visit
Visit ──(1:N)──► Lab Result
Enrollment ──(1:N)──► Adverse Event
Patient ──(1:N)──► Prescription
Patient ──(1:N)──► Diagnosis
Patient ──(1:N)──► Medical Procedure
Enrollment ──(1:N)──► Consent
Prescription ──(N:1)──► Drug
Prescription ──(N:1)──► Physician
Trial Site ──(N:1)──► Physician (PI)
Patient ──(N:1)──► Insurance Plan
```

Cross-Entity Analytics

- **Patient 360:** Patient → Enrollment → Visit → Lab Result + Adverse Event + Prescription + Diagnosis
- **Trial Overview:** Clinical Trial → Trial Site → Enrollment → Visit
- **Drug Safety:** Drug → Prescription → Patient → Adverse Event
- **Site Performance:** Hospital → Trial Site → Enrollment (completion rates)

This document supports ontology mapping and knowledge graph construction using OntoBricks. The data model and relationships described above are designed to be directly translatable into RDF/OWL ontology structures for semantic querying and AI-driven insights.